

		$(D) \text{ Total KE} = \frac{2(3D)}{2} + \frac{2D}{2} = D$
12	D	Same temperature implies thermal equilibrium. Hence all 3 objects are in thermal equilibrium and have the same temperature. However the heat capacity of each material is different, so the amount of internal energy is different. Objects in thermal equilibrium can exchange thermal energy, but there will be no net exchange of thermal energy.
13	B	As n and T are constant